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Geography

Resources and Development (India: People and Economy)

Land Resources, Agricultural Systems, Water Dynamics, Mineral Belts, Energy Transition & Major Industrial Corridors

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Module 1: Land Resources & Agricultural Land Use

1.1 Land-Use Records and 9-Fold Reporting Categories

Land is the most fundamental natural resource supporting all terrestrial ecosystems, agricultural production, and human infrastructure. In India, official land-use data is recorded by state land revenue departments. The Ministry of Agriculture and Farmers Welfare classifies the country's total reporting area into **9 distinct land-use categories**:

THE 9-FOLD LAND CLASSIFICATION SYSTEM IN INDIA

1. **Forests:** Area demarcated and notified as forest land under statutory legislation, irrespective of the actual crown density or vegetative canopy cover present.
2. **Area under Non-Agricultural Uses:** Land occupied by human settlements (rural and urban dwellings), transport infrastructure (roads, railways, canals), industries, and civic utilities.
3. **Barren and Unculturable Waste Land:** Barren rocky outcrops, steep ravines, hyper-arid deserts, and mountain screes that cannot be brought under cultivation without prohibitively high capital expenditure.
4. **Permanent Pastures and Grazing Lands:** Common village grazing lands (Panchayat pastures) dedicated to livestock feeding.
5. **Area under Miscellaneous Tree Crops and Groves:** Land under casuarina, thatching grasses, bamboo clumps, and fruit orchards not included in the Net Sown Area.
6. **Culturable Waste Land:** Land capable of being cultivated but left uncultivated for **more than 5 consecutive years**. It can be rehabilitated through soil reclamation.
7. **Fallow Other than Current Fallow:** Agricultural land left uncultivated for an intermediate period of **1 to 5 years** to replenish organic soil fertility.
8. **Current Fallow:** Cropland left unseeded and fallowed for **1 agricultural year or less** during the reporting cycle.
9. **Net Sown Area (NSA):** Physical terrestrial land area on which crops are sown and harvested at least once during an agricultural year.

1.2 Temporal Changes in Land Use & Cropping Intensity

Over the six decades of developmental planning since 1950–51, India's land-use patterns have undergone significant structural shifts:

- **Expansion of Non-Agricultural Land:** Rapid urban expansion, industrialization, expressways, and infrastructure development have driven the largest percentage increase in land devoted to non-agricultural uses.
- **Stabilization of Net Sown Area:** The Net Sown Area expanded rapidly during early green revolution decades but has plateaued around 140 million hectares (45.5% of total reporting area), indicating that physical limits to horizontal land expansion have been reached.
- **Decline in Pastures & Common Lands:** Population pressure has caused encroachment and privatization of common property resources (CPRs), creating severe shortages of cattle fodder.

CRUCIAL MEASURES OF AGRICULTURAL LAND UTILIZATION

$$\text{Gross Cropped Area (GCA)} = \text{Net Sown Area} + \text{Area Sown More Than Once in a Single Year}$$

$$\text{Cropping Intensity} = \left(\frac{\text{Gross Cropped Area (GCA)}}{\text{Net Sown Area (NSA)}}\right) \times 100$$

- In states with high irrigation density (Punjab, Haryana, West Bengal), cropping intensity exceeds **150% to 190%**, whereas in rainfed dryland zones (Rajasthan, Gujarat), cropping intensity remains below **110% to 120%**.

1.3 The Three Distinct Cropping Seasons

India's agricultural year spans from June to May and is divided into three distinct agro-climatic cropping seasons:

- **1. Kharif Season (June to September):** Coincides with the arrival and progression of the Southwest Monsoon. Major crops include **Rice, Cotton, Jute, Jowar, Bajra, Maize, Tur (Arhar), Groundnut, and Soyabean**.
- **2. Rabi Season (October to March):** Begins with the onset of autumn/winter and harvested in early spring. Governed by cool, dry weather and winter cyclonic depressions (Western Disturbances) in the north. Major crops include **Wheat, Gram, Rapeseed and Mustard, Barley, and Peas**.
- **3. Zaid Season (April to June):** A short summer cropping cycle between the harvest of Rabi crops and the arrival of the Kharif monsoon. Cultivated under assured irrigation, featuring **Watermelon, Muskmelon, Cucumber, Gourds, Vegetables, and green fodder**.

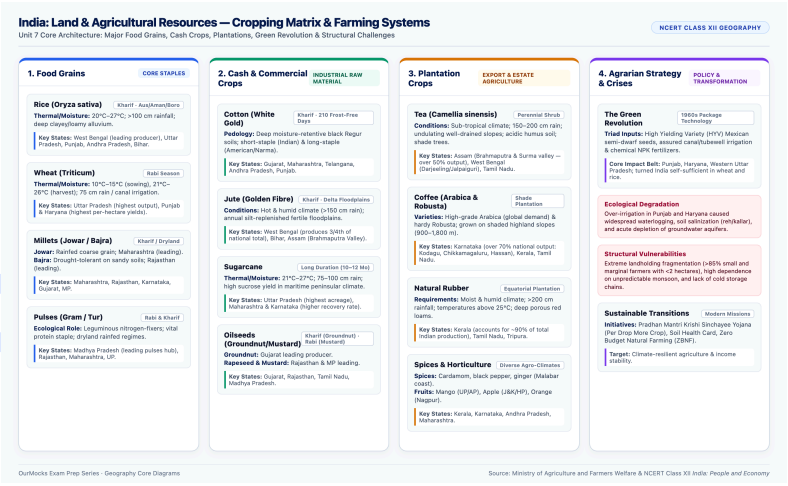


Figure 1: India: Land & Agricultural Resources — Cropping Matrix, Agro-Ecological Zones & Farming Challenges

Comprehensive Classification of Land Utilization in India

Land Category	Temporal Trend (1950–2015+)	Key Geographic Dynamics & Drivers
Net Sown Area (NSA)	Initial expansion, now stabilized 45.5%	Physical ceiling reached; future output relies exclusively on cropping intensity and yields.
Non-Agricultural Uses	Massive continuous expansion	Driven by urbanization, highway networks, industrial parks, and rural settlement growth.
Forest Cover	Gradual statutory increase (23.3%)	Still below the 33% ecological threshold mandated by the National Forest Policy 1988.
Permanent Pastures	Steady decline and fragmentation	Pressured by agrarian encroachment; severe reduction in common property grazing land.
Culturable Waste	Gradual decline due to reclamation	Brought under cultivation through canal colonization and wasteland reclamation schemes.

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Module 2: Major Agricultural Crops & Spatial Distribution

2.1 Food Grains: Staples, Millets & Pulses

Food grains occupy approximately two-thirds of the total cropped area in India, anchoring national food security:

STAPLE CEREALS: RICE AND WHEAT

- **Rice (Oryza sativa):**
  - **Agro-Climatic Requirements:** High temperature (20°C–27°C), high humidity, and annual rainfall exceeding **100 cm**. Cultivated on deep, moisture-retentive clayey and loamy alluvial soils.
  - **Seasonal Varieties:** In eastern and southern states (West Bengal, Assam, Odisha, Tamil Nadu), three rice crops are cultivated annually: **Aus** (autumn crop), **Aman** (winter crop — largest share), and **Boro** (summer crop).
  - **Leading Producing States: West Bengal** (leading national producer), **Uttar Pradesh**, **Punjab**, **Andhra Pradesh**, **Bihar**, and **Tamil Nadu**. Punjab and Haryana achieve the highest per-hectare yields via tubewell canal irrigation.
- **Wheat (Triticum):**
  - **Agro-Climatic Requirements:** Cool growing season with temperatures of 10°C–15°C at sowing and warm sunny weather (21°C–26°C) during harvest. Requires 75 cm rainfall or canal irrigation.
  - **Leading Producing States: Uttar Pradesh** (highest aggregate production), **Punjab** (highest per-hectare yield exceeding 4,500 kg/ha), **Haryana**, **Madhya Pradesh**, and **Rajasthan**.

- **Coarse Cereals & Millets:**
  - **Jowar (Sorghum):** A rainfed crop cultivated in semi-arid tracts; accounts for 5% of cropped area. **Maharashtra** is the undisputed leader, followed by Karnataka, Madhya Pradesh, and Andhra Pradesh.
  - **Bajra (Pearl Millet):** Exceptionally drought-resistant; grown on sandy and shallow black soils. **Rajasthan** is the leading producer, followed by Uttar Pradesh, Gujarat, and Haryana.
  - **Maize:** Grown as both food and poultry feed across diverse agro-climatic zones. Leading producers include **Karnataka**, **Madhya Pradesh**, **Bihar**, and **Maharashtra**.
- **Pulses (Gram and Tur/Arhar):**
  - Leguminous crops that restore soil fertility by biologically fixing atmospheric nitrogen (except arhar/tur). Concentrated in dryland rainfed regimes.
  - **Madhya Pradesh** is the leading pulse-producing state in India, followed by Rajasthan, Maharashtra, Uttar Pradesh, and Karnataka.

2.2 Commercial Cash Crops, Fibres & Oilseeds

- **Cotton (Fibre Crop):**
  - Requires high temperature, light rainfall or irrigation, **210 frost-free days**, and bright sunshine. Flourishes in the moisture-retentive deep black **Regur soils** of the Deccan plateau.
  - Two broad varieties: Short-staple (indigenous Indian cotton) and Long-staple (American cotton, known as **Narma** in northwestern India).
  - Leading producers: **Gujarat**, **Maharashtra**, **Telangana**, **Andhra Pradesh**, and **Punjab**.

- **Jute (Golden Fibre):**
  - Requires high temperatures (>25°C), heavy rainfall (>150 cm), and fertile, silt-replenished floodplains.
  - Concentrated in the eastern deltas: **West Bengal** produces nearly three-fourths of India's aggregate jute output, followed by Bihar and Assam (Brahmaputra Valley).
- **Sugarcane:**
  - A tropical and subtropical perennial grass requiring temperatures of 21°C–27°C and 75–100 cm rainfall.
  - **Uttar Pradesh** has the largest acreage and highest aggregate production. However, peninsular states (**Maharashtra**, **Karnataka**, **Tamil Nadu**) achieve significantly higher sucrose content and per-hectare yields due to their maritime climate and longer crushing seasons.
- **Oilseeds:**
  - **Groundnut:** A Kharif legume; **Gujarat** is the leading producer (30% of national output), followed by Rajasthan, Tamil Nadu, and Andhra Pradesh.
  - **Rapeseed and Mustard:** Subtropical Rabi oilseeds grown in dry north/central belts; **Rajasthan** is the premier producer.

2.3 Plantation Crops: Tea and Coffee

- **Tea (Camellia sinensis):** Indigenous to the hill slopes of northern Myanmar and Assam. Requires sub-tropical climate, high annual rainfall (150–200 cm), undulating well-drained mountain slopes, acidic humus-rich soil, and shade trees. **Assam** produces over 50% of India's tea, followed by West Bengal (Darjeeling and Jalpaiguri districts) and the Nilgiri hills in Tamil Nadu.
- **Coffee:** Grown under shaded canopies at altitudes of 900–1,800 meters. India predominantly cultivates high-grade **Arabica** (prized globally) and **Robusta**. **Karnataka** is the dominant producer (accounting for over 70% of national output, centered in Kodagu, Chikkamagaluru, and Hassan districts), followed by Kerala and Tamil Nadu.

Agro-Climatic Requirements of Major Indian Crops

Crop	Thermal & Moisture	Soil & Topography	Top Producing States
Rice	20°C–27°C; >100 cm rain	Deep clayey/loamy alluvium; flat plains	West Bengal, Uttar Pradesh, Punjab, AP.
Wheat	10°C–15°C (sow), 21°C–26°C (harv)	Well-drained loams; plains with canal water	Uttar Pradesh, Punjab, Haryana, MP.
Cotton	21°C–30°C; 210 frost-free days	Deep moisture-retentive black Regur soil	Gujarat, Maharashtra, Telangana, Punjab.
Jute	>25°C; >150 cm humidity	Rich silt-replenished river floodplains	West Bengal (3/4th total), Bihar, Assam.
Tea	20°C–30°C; 150–200 cm rain	Well-drained undulating acidic hill slopes	Assam (Brahmaputra Valley), WB, TN.
Coffee	15°C–28°C; 150–200 cm rain	Highland shaded slopes; well-drained loams	Karnataka (Kodagu, Hassan), Kerala, TN.

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Module 3: Agricultural Development & Structural Crises

3.1 The Green Revolution: Strategy and Impact

At independence in 1947, Indian agriculture was characterized by extreme feudal stagnation, famine vulnerability, and precarious dependence on food imports (under the US PL-480 grain assistance program). In the mid-1960s, the Government of India launched the historic **Green Revolution** strategy:

THE GREEN REVOLUTION PACKAGE TECHNOLOGY

The Green Revolution replaced traditional farming with a modern technological package anchored by three core inputs:

1. **High Yielding Varieties (HYV Seeds):** Introduction of semi-dwarf, fertilizer-responsive Mexican wheat varieties (developed by Dr. Norman Borlaug) and IR-8 miracle rice strains.
2. **Assured Irrigation Infrastructure:** Extensive development of perennial canal systems (Bhakra-Nangal, Western Yamuna Canal) and energized tubewells.
3. **Chemical Fertilizers & Agrochemicals:** Application of high doses of chemical nitrogen, phosphorus, and potassium (NPK) fertilizers, pesticides, and weedicides.

- **Spatial Core:** Initially restricted to regions with assured irrigation: **Punjab, Haryana, and Western Uttar Pradesh**.
- **National Outcome:** India achieved national food self-sufficiency, transformed from a "ship-to-mouth" economy into a net grain exporter, and created substantial grain reserves managed by the Food Corporation of India (FCI).

3.2 Ecological Fallout & Second-Generation Crises

Despite achieving self-sufficiency in wheat and rice, the Green Revolution spawned profound environmental and socio-economic crises:

- **Soil Salinization & Waterlogging:** Over-irrigation in poorly drained canal commands of Punjab, Haryana, and Rajasthan caused rising water tables, bringing underground mineral salts to the topsoil via capillary action, rendering vast acres barren with saline encrustations (locally called **Reh**, **Kallar**, or **Usar**).
- **Severe Depletion of Groundwater Aquifers:** Intensive tubewell exploitation for water-guzzling crops (paddy cultivation in semi-arid Punjab) caused groundwater tables to plunge by several meters, turning dark-zone blocks critically over-exploited.
- **Nutrient Imbalance & Chemical Toxicity:** Excessive, unbalanced use of subsidized urea disrupted natural soil chemistry, destroyed beneficial soil microflora, and contaminated rural drinking water sources with nitrates and heavy metal residues.
- **Inter-Regional & Class Disparities:** The early green revolution heavily favored large landowning farmers who could afford mechanized tractors and tubewells, widening economic cleavages between the irrigated northwestern plains and rainfed eastern/central dryland states.

3.3 Structural Challenges of Indian Agriculture

Indian agriculture continues to confront chronic structural handicaps:

- **Fragmentation of Landholdings:** Due to inheritance laws and demographic explosion, the average farm holding size has shrunk to **1.08 hectares**. Over **86% of all operational holdings belong to small and marginal farmers** (under 2 hectares), hindering farm mechanization and economies of scale.

- **Monsoon Gamble:** Despite massive irrigation schemes, over 50% of the Net Sown Area remains purely rainfed, leaving crop yields vulnerable to erratic monsoon droughts and El Niño weather cycles.
- **Inadequate Rural Credit & Distress:** Exploitation by informal village moneylenders charging usurious interest rates, volatile agricultural market price crashes during bumper harvests, and lack of climate-controlled cold storage chains fuel persistent agrarian distress.

Key Government Missions for Sustainable Agriculture

Mission Name	Nodal Scheme	Strategic Objectives & Field Interventions
PM-KSY (Krishi Sinchayee Yojana)	Ministry of Agriculture & Jal Shakti	Expand cultivated area with assured irrigation; promote micro-irrigation under **"Per Drop More Crop"** (drip & sprinkler systems).
Soil Health Card Scheme	Ministry of Agriculture	Periodic laboratory testing of topsoil nutrients; advising farmers on customized NPK and micronutrient dosage to prevent soil degradation.
PKVY & Natural Farming (Paramparagat Krishi Vikas)	Ministry of Agriculture	Promote certified organic farming and Zero Budget Natural Farming (ZBNF) to eliminate chemical fertilizers.
PM-FASAL Bima Yojana	Ministry of Agriculture	Comprehensive multi-peril crop insurance covering non-preventable natural risks from pre-sowing to post-harvest stages.

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Module 4: Water Resources: Availability, Dynamics & Conservation

4.1 Water Availability and Spatial Distribution

Water is a cyclic, renewable resource, yet its spatial and temporal availability across India is profoundly uneven due to monsoonal seasonality:

- **Surface Water Resources:** India receives approximately 4,000 billion cubic meters (BCM) of precipitation annually. The aggregate surface runoff in river basins is estimated at **1,869 BCM**. However, due to severe topographical, hydrological, and geological constraints, **only 690 BCM (approx. 32%) is utilisable**.
  - **River Basin Disparities:** The **Ganga and Brahmaputra basins** together account for nearly **60% of the country's total surface water discharge**, while river basins in peninsular India (Godavari, Krishna, Mahanadi, Kaveri) carry seasonal, fluctuating flows.
- **Replenishable Groundwater Resources:** The total annual replenishable groundwater potential is estimated at **433 BCM**. The vast alluvial plains of northern India possess immense groundwater reservoirs, whereas the hard crystalline rock terrain of Peninsular India has restricted storage.

SECTORAL WATER CONSUMPTION PATTERN IN INDIA

In India, water consumption is overwhelmingly tilted toward the primary sector:

- **Surface Water Withdrawal:** Agriculture accounts for **89%**, Domestic sector accounts for **9%**, and Industrial sector accounts for **2%**.
- **Groundwater Withdrawal:** Agriculture accounts for **92%**, Domestic sector accounts for **5%**, and Industrial sector accounts for **3%**.
- Rising urbanization and rapid industrial manufacturing are projected to substantially escalate domestic and industrial water demand in coming decades.

4.2 Spatial Disparities in Groundwater Exploitation

The **Stage of Groundwater Development** (evaluated as the ratio of annual groundwater draft to annual replenishable recharge) varies starkly across Indian states:

- **Over-Exploited Zones (>100% Development): Punjab (166%)**, **Haryana (134%)**, **Rajasthan (140%)**, and **Delhi**. In these states, annual groundwater extraction far exceeds natural annual recharge, causing catastrophic water table collapse and drying of shallow wells.
- **Critical and Semi-Critical Zones (70%–100%):** Gujarat, Uttar Pradesh, Tamil Nadu, and parts of Maharashtra.
- **Under-Exploited / High-Potential Zones (under 50%):** Eastern states (**Bihar, West Bengal, Odisha, Assam, Chhattisgarh**). These regions have abundant unutilized groundwater potential that can support sustainable agricultural expansion through regulated tubewells.

4.3 Watershed Management & Rainwater Harvesting

To combat worsening hydrological crises, decentralized watershed management has emerged as a national priority:

PIONEERING WATERSHED MANAGEMENT INITIATIVES

Watershed management entails the rational, sustainable utilization and conservation of surface and groundwater resources alongside land, soil, and vegetation within a natural drainage basin.

- **Haryali:** A flagship watershed development program sponsored by the Central Government, executed directly by Gram Panchayats with people's active participation to enable rural drinking water, irrigation, fisheries, and afforestation.
- **Neeru-Meeru (Water and You):** Launched by the Government of Andhra Pradesh; focused on community excavation of check-dams, percolation tanks, and recharge trenches to replenish groundwater tables.
- **Arvary Pani Sansad:** A pioneering community-led water parliament initiated by rural communities with NGO support in Alwar, Rajasthan, constructing small earthen check-dams (**Johads**) that revived dead perennial streams.
- **Statutory Rainwater Harvesting (RWH): Tamil Nadu** became the first state in India to make rooftop rainwater harvesting **legally mandatory** across all existing and newly constructed buildings, drastically recharging local water tables.

Comparative Stages of Groundwater Development in Indian States

Category	Development Stage	Representative States & Environmental Consequences
Over-Exploited	Above 100% extraction	**Punjab**, **Haryana**, **Rajasthan**, **Delhi**. Severe water table plunge, salinity.

Category	Development Stage	Representative States & Environmental Consequences
Critical / High	70% to 100% extraction	**Gujarat**, **Tamil Nadu**, **Uttar Pradesh**. Fluoride leaching in aquifers.
Moderate	50% to 70% extraction	**Karnataka**, **Maharashtra**, **Andhra Pradesh**, **Madhya Pradesh**.
Under-Exploited	Below 50% extraction	**Bihar**, **West Bengal**, **Odisha**, **Assam**. Immense replenishable potential.

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Module 5: Mineral & Energy Resources: Belts & Transition

5.1 Classification and Spatial Mineral Belts

Minerals are natural inorganic substances possessing definite chemical compositions and physical crystalline structures. India is richly endowed with metallic and non-metallic minerals, predominantly concentrated within the ancient crystalline shield of the Peninsular plateau:

THE THREE PRINCIPAL MINERAL BELTS OF INDIA

- **1. The North-Eastern Plateau Region:**
  - Encompasses the Chotanagpur Plateau (Jharkhand), Odisha plateau, and parts of West Bengal and Chhattisgarh.
  - Termed the **"Ruhr of India"** due to immense concentrations of high-grade **Iron ore, Coking Coal, Manganese, Bauxite, and Mica**.
- **2. The South-Western Plateau Region:**
  - Extends across Karnataka, Goa, and contiguous upland Tamil Nadu and Kerala.
  - Rich in high-grade **Iron ore** (Ballari-Hospet, Kudremukh, Goa), **Manganese**, **Limestone**, and **Bauxite**. It lacks fossil coal reserves, but possesses massive **Lignite (brown coal)** at Neyveli (Tamil Nadu) and rich **Monazite sands** (Thorium) on the Kerala coast.
- **3. The North-Western Region:**
  - Spans the Aravalli range in Rajasthan and adjoining parts of Gujarat.
  - Specialized in non-ferrous minerals: **Copper** (Khetri), **Zinc and Lead** (Zawar mines, Udaipur), **Marble, Granite, Gypsum**, and **Fuller's earth**. Gujarat possesses rich **Petroleum and Natural Gas** reserves in the Cambay basin and salt pans along the coast.

5.2 Metallic and Non-Metallic Minerals

- **Ferrous Minerals (Iron Ore and Manganese):**
  - **Iron Ore:** India possesses world-class reserves of **Hematite** (60–70% iron content) and **Magnetite** (high magnetic properties). Key producing belts:
    1. **Odisha-Jharkhand Belt:** High-grade hematite mined at **Badampahar** (Mayurbhanj), **Kendujhar**, **Noamundi**, and **Gua** (Singhbhum).
    2. **Durg-Bastar-Chandrapur Belt:** Super-high grade hematite at **Bailadila** range (Bastar, Chhattisgarh), exported to Japan via Visakhapatnam port.

3. **Ballari-Chitradurga-Chikkamagaluru Belt: Kudremukh** (100% export unit transported as slurry pipeline to Mangaluru port) and Ballari-Hospet.

- **Manganese:** Essential for smelting steel (approx. 10 kg required per tonne of steel). Leading producers: **Odisha** (Sundargarh, Kendujhar), **Madhya Pradesh** (Balaghat), and **Maharashtra** (Nagpur, Bhandara).

- **Non-Ferrous Minerals (Bauxite and Copper):**
  - **Bauxite (Aluminium Ore):** Formed by tropical weathering of rocks rich in aluminium silicates. **Odisha** is the leading producer (accounting for over 50% of national output), centered at **Panchpatmali in Koraput district**, Kalahandi, and Bolangir. Other fields: Amarkantak plateau (MP) and Katni-Jabalpur.
  - **Copper:** Critically deficient in India. Primary mines: **Balaghat** (MP — over 50% of output), **Khetri** (Rajasthan), and **Singhbhum** (Jharkhand).
- **Non-Metallic Minerals (Mica):**
  - Outstanding dielectric strength and insulating resistance for electronic/electrical industries. Leading belt: **Jharkhand** (Koderma-Hazaribagh belt — Koderma is known as the "Mica Capital of India"), followed by **Nellore** (Andhra Pradesh) and Rajasthan.

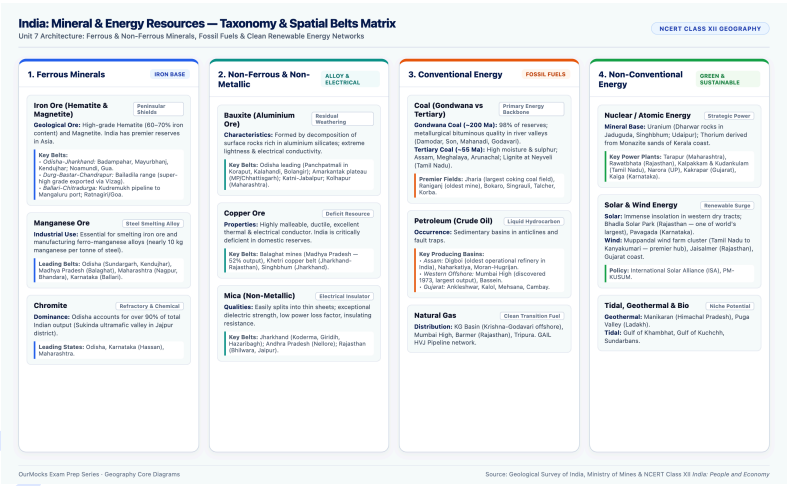


Figure 2: India: Mineral & Energy Resources — Taxonomy, Spatial Belts & Conventional/Renewable Energy Networks

5.3 Conventional and Non-Conventional Energy

- **Coal:** Primary commercial energy source supplying over 55% of India's electricity:
  - **Gondwana Coal (200 million years old):** Accounts for **98% of total coal reserves** and almost all metallurgical coking coal. Distributed along river valleys: **Damodar Valley** (Jharia — largest coking coal field; Raniganj — oldest coal mine; Bokaro), Son, Mahanadi (Talcher), and Godavari (Singareni).
  - **Tertiary Coal (55 million years old):** High moisture and sulfur content; found in Assam (Makum), Meghalaya (Cherrapunji, Daranggiri), and Jammu & Kashmir. Massive **Lignite** deposits are mined at **Neyveli** (Tamil Nadu) for thermal power.
- **Petroleum & Natural Gas:**
  - Occurs in anticlinal traps of sedimentary rock strata. Key basins: **Digboi** (Assam — oldest operational oilfield and refinery in India, 1901), **Naharkatiya**, **Mumbai High** (discovered 1973, located 160 km offshore, largest domestic producer), **Bassein**, and **Ankleshwar** (Gujarat).
  - Natural gas is transported via the landmark **HVJ Pipeline** (Hazira-Vijaipur-Jagdishpur, 1,750 km) operated by GAIL.
- **Non-Conventional & Clean Energy:**
  - **Nuclear Energy:** Uses **Uranium** (Dharwar rocks in Jaduguda, Jharkhand; Aravallis in Rajasthan) and **Thorium** (derived from Monazite beach sands of Kerala). Key reactors:

Tarapur (Maharashtra — first nuclear plant), **Rawatbhata** (Rajasthan), **Kudankulam** and **Kalpakkam** (Tamil Nadu), **Narora** (UP), **Kakrapar** (Gujarat), and **Kaiga** (Karnataka).

- **Solar & Wind Power:** Immense solar insolation in Western Rajasthan (**Bhadla Solar Park** — one of the world's largest installations). Wind energy is led by **Tamil Nadu** (**Muppandal wind farm** cluster near Kanyakumari) and Rajasthan (Jaisalmer).

Master Mineral and Energy Matrix of India

Resource	Geological Era / Type	Premier Mining Centers	Leading Producing States
Iron Ore	Hematite / Magnetite	Badampahar, Noamundi, Bailadila, Kudremukh	Odisha, Chhattisgarh, Karnataka, Jharkhand.
Manganese	Ferrous smelting alloy	Balaghat, Sundargarh, Nagpur-Bhandara	Madhya Pradesh, Odisha, Maharashtra.
Bauxite	Aluminium ore	Panchpatmali (Koraput), Amarkantak, Katni	Odisha (>50% national total), MP, Gujarat.
Coal	Gondwana (Bituminous)	Jharia, Raniganj, Bokaro, Talcher, Korba	Jharkhand, Odisha, Chhattisgarh, WB.
Petroleum	Sedimentary Tertiary	Digboi, Mumbai High, Ankleshwar, Kalol	Offshore (Mumbai High), Gujarat, Assam.
Nuclear	Uranium & Thorium	Jaduguda (Uranium), Kerala coast (Monazite)	Tarapur, Rawatbhata, Kudankulam, Kaiga.

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Module 6: Manufacturing Industries, Spatial Corridors & Unit Synthesis

6.1 Factors of Industrial Location & Major Corridors

Industrial location is determined by complex spatial interactions of raw material availability, power, market access, transport costs, labor supply, and historical colonial agglomeration:

- **Raw Material-Tied Industries:** Industries using heavy, bulky, weight-losing raw materials (e.g., Iron & Steel, Sugar, Cement) are located immediately adjacent to raw material sources.
- **Market-Tied Industries:** Industries producing perishable or fragile commodities, or utilizing ubiquitous inputs (e.g., bakeries, electronics, petroleum refineries processing imported crude like Panipat, Mathura, and Koyali) locate near consumption markets.

THE 8 MAJOR INDUSTRIAL REGIONS OF INDIA

1. **Mumbai-Pune Industrial Region:** Textile mills, petrochemicals, oil refineries, engineering goods, software, and pharmaceuticals.
2. **Hugli Industrial Region:** Extends along the Hugli river; anchored by jute manufacturing, tea processing, engineering, cotton textiles, and river transport.
3. **Bengaluru-Tamil Nadu Industrial Region:** Heavy public sector electronics, aerospace (HAL), automobiles, machine tools, and cotton textiles (Coimbatore).
4. **Gujarat Industrial Region:** Centered around Ahmedabad and Vadodara; petrochemical complexes, cotton textiles, diamond cutting, pharmaceuticals, and chemicals.

5. **Chotanagpur Industrial Region:** The heavy metallurgical spine of India; iron and steel plants (Jamshedpur, Bokaro, Durgapur), coalfields, and heavy machinery (Ranchi).
6. **Vishakhapatnam-Guntur Industrial Region:** Port-based steel plant (VSP), oil refining, shipbuilding, fertilizer, and petrochemical industries.
7. **Gurgaon-Delhi-Meerut Industrial Region:** High-growth electronics, automobiles, software, light engineering, agro-processing, and consumer durables.
8. **Kollam-Thiruvananthapuram Industrial Region:** Plantation-based processing (coir, rubber, cashew), food processing, and marine fish canning.

6.2 Key Manufacturing Industries in India

- **Iron and Steel Industry:**
  - **TISCO (Tata Iron and Steel Company):** Established at Sakchi (Jamshedpur) in 1907; utilizes hematite from Noamundi/Gurumahisani, coking coal from Jharia, limestone from Birmitrapur, and water from Subarnarekha river.
  - **Public Sector Plants (SAIL): Bhilai** (Chhattisgarh, Russian collaboration, iron ore from Dalli-Rajhara), **Rourkela** (Odisha, German collaboration), **Durgapur** (West Bengal, British collaboration), and **Bokaro** (Jharkhand, Russian collaboration). **Visakhapatnam Steel Plant (Rashtriya Ispat Nigam Ltd - RINL)** is India's premier shore-based steel complex.
- **Cotton Textile Industry:** India's oldest organized manufacturing sector. Concentrated initially in Mumbai and Ahmedabad (raw cotton, port, humid climate); decentralized post-independence to Coimbatore, Kanpur, Ludhiana, and Surat.
- **Sugar Industry:** Seasonal weight-losing agro-industry. Underwent a major geographic shift from UP/Bihar to peninsular states (**Maharashtra, Karnataka, Tamil Nadu**) due to higher sucrose recovery and longer crushing seasons.

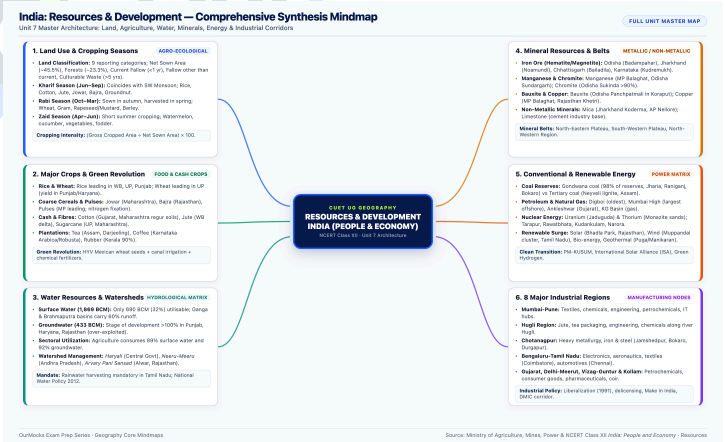


Figure 3: India: Resources & Development — Comprehensive Synthesis Mindmap of Agriculture, Minerals & Industrial Hubs

Master Synthesis Matrix of Resources and Development in India

Resource Sector	Core Structural Attributes	Key Spatial Hallmark & Policy Mandate
Land Resources	9-fold classification; NSA 45.5%	Physical land ceiling reached; output depends on cropping intensity (GCA/NSA) and Soil Health Cards.

Resource Sector	Core Structural Attributes	Key Spatial Hallmark & Policy Mandate
Agriculture	Food grains, cash crops, plantations	Green Revolution achieved food security; contemporary focus on PM-KSY, PKVY, and climate resilience.
Water Dynamics	Surface (690 BCM) & Ground (433 BCM)	Agriculture consumes >90% water; groundwater over-exploited in NW; Haryali and RWH watershed priorities.
Mineral Belts	NE Plateau, SW Plateau, NW Region	Odisha-Jharkhand iron ore; Bailadila export hematite; Koraput bauxite; Koderma mica.
Energy Resources	Gondwana coal, offshore oil, renewables	Gondwana coal supplies 98% reserves; rapid transition to solar (Bhadla) and wind (Muppandal).
Industrial Regions	8 major manufacturing corridors	Chotanagpur metallurgical core; Mumbai-Pune commercial hub; 1991 liberalization and DMIC corridor.

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OurMocks

Module 7: CUET High-Yield Practice Questions

Q1. Under the 9-fold land-use classification in India, what is the term used for cultivable land that has been left fallow and unseeded for more than five consecutive years?

(A) Current fallow
(B) Fallow other than current fallow
(C) Barren and unculturable waste
(D) Culturable waste land

Sol. Q1. Culturable waste land is land capable of being cultivated but left fallowed for more than 5 consecutive years. It can be rehabilitated through soil reclamation.

Answer: (D)

Q2. In the context of Indian agriculture, which of the following mathematical formulas accurately represents 'Cropping Intensity'?

(A) $\left(\frac{\text{Gross Cropped Area}}{\text{Net Sown Area}}\right) \times 100$
(B) $\left(\frac{\text{Gross Cropped Area}}{\text{Total Geographical Area}}\right) \times 100$
(C) $\left(\frac{\text{Net Sown Area}}{\text{Gross Cropped Area}}\right) \times 100$
(D) $\left(\frac{\text{Net Cultivated Area}}{\text{Total Agricultural Population}}\right) \times 100$

Sol. Q2. Cropping Intensity is computed as (Gross Cropped Area ÷ Net Sown Area) × 100, measuring the frequency of cropping on net sown land within a single year.

Answer: (A)

Q3. In eastern Indian states such as West Bengal and Assam, farmers cultivate three seasonal crops of rice in an agricultural year. These three crops are locally designated as:

(A) Aus, Aman, and Boro
(B) Rabi, Kharif, and Zaid
(C) Jowar, Bajra, and Ragi
(D) Kallar, Reh, and Usar

Sol. Q3. In West Bengal and Assam, the three seasonal rice crops grown across the agricultural cycle are known as Aus (autumn), Aman (winter), and Boro (summer).

Answer: (A)

Q4. Which state is the undisputed leader in coffee production in India, accounting for over 70% of total national output from districts like Kodagu and Chikkamagaluru?

(A) Kerala
(B) Tamil Nadu
(C) Karnataka
(D) Andhra Pradesh

Sol. Q4. Karnataka produces over 70% of India's aggregate coffee output, primarily cultivating premium Arabica and Robusta varieties in the Western Ghats highlands.

Answer: (C)

Q5. Which ecological problem emerged as a severe consequence of excessive canal over-irrigation in the Green Revolution heartlands of Punjab and Haryana?

(A) Arsenic contamination in shallow wells
(B) Severe loss of organic topsoil due to wind deflation
(C) Accelerated laterization of alluvial soils
(D) Soil salinization, waterlogging, and formation of 'kallar' encrustations

Sol. Q5. Over-irrigation in poorly drained canal commands caused water tables to rise, bringing mineral salts to the surface via capillary action, creating saline kallar soils.

Answer: (D)

Q6. Approximately what percentage of the total annual replenishable surface water resources in India (1,869 BCM) is estimated to be practically utilisable?

(A) Approx. 32% (690 BCM)
(B) Approx. 58% (1,080 BCM)
(C) Approx. 75% (1,400 BCM)
(D) Approx. 90% (1,680 BCM)

Sol. Q6. Out of 1,869 BCM of aggregate surface water flow, only 690 BCM (approx. 32%) is utilisable due to physiological, relief, and hydrological limitations.

Answer: (A)

Q7. Which sector accounts for the largest share of both surface water withdrawals (89%) and groundwater withdrawals (92%) in India?

(A) Domestic municipal sector
(B) Thermal power generation
(C) Agricultural irrigation
(D) Industrial manufacturing

Sol. Q7. Agriculture is the overwhelmingly dominant consumer of water in India, utilizing 89% of surface water and 92% of groundwater extractions.

Answer: (C)

Q8. Which was the first state in India to make rooftop rainwater harvesting (RWH) legally mandatory for all buildings across the state?

(A) Rajasthan
(B) Maharashtra
(C) Tamil Nadu
(D) Gujarat

Sol. Q8. Tamil Nadu pioneered mandatory rooftop rainwater harvesting across all residential and commercial buildings, reversing severe water table decline.

Answer: (C)

Q9. The community-based watershed management parliament 'Arvary Pani Sansad', which successfully revived dried-up seasonal streams using check-dams, originated in:

(A) Ralegan Siddhi, Maharashtra
(B) Alwar district, Rajasthan
(C) Jhabua district, Madhya Pradesh
(D) Koraput district, Odisha

Sol. Q9. The Arvary Pani Sansad was established in Alwar, Rajasthan, constructing johads (earthen check-dams) to revitalize seasonal river basins.

Answer: (B)

Q10. The world-famous 'Bailadila' iron ore range, celebrated for producing super-high-grade hematite ore that is exported to Japan, is located in:

(A) Mayurbhanj district, Odisha
(B) Bastar district, Chhattisgarh
(C) Singhbhum district, Jharkhand
(D) Ballari district, Karnataka

Sol. Q10. The Bailadila iron ore range is located in the Bastar district of Chhattisgarh. Its high-grade hematite is transported to Visakhapatnam port for export.

Answer: (B)

Q11. Which mine in Odisha is the single largest bauxite-producing deposit in India, supplying the aluminium smelting complex of NALCO?

(A) Panchpatmali in Koraput district
(B) Badampahar in Mayurbhanj district
(C) Sukinda in Jajpur district
(D) Noamundi in Kendujhar district

Sol. Q11. Panchpatmali deposits in Koraput district of Odisha are the premier bauxite mining fields in India, feeding the Damanjodi refinery of NALCO.

Answer: (A)

Q12. Gondwana coal formations account for approximately what proportion of India's total coal reserves and commercial production?

(A) 50%
(B) 75%
(C) 85%
(D) 98%

Sol. Q12. Gondwana coal fields (located in Damodar, Son, Mahanadi, and Godavari river basins) account for 98% of India's total coal reserves and all coking coal.

Answer: (D)

Q13. Which is the oldest operational oilfield and refinery in India, having begun commercial operations as early as 1901?

(A) Mumbai High
(B) Ankleshwar
(C) Digboi
(D) Kalol

Sol. Q13. Digboi in Assam is India's and Asia's oldest operational commercial oilfield and refinery, commissioned in 1901 in the Brahmaputra valley.

Answer: (C)

Q14. The world-renowned 'Muppandal' wind farm cluster, representing one of the largest onshore wind power installations in India, is located in:

(A) Jaisalmer, Rajasthan
(B) Kanyakumari, Tamil Nadu
(C) Kutch, Gujarat
(D) Satara, Maharashtra

Sol. Q14. The Muppandal wind farm cluster is situated in Kanyakumari district, Tamil Nadu, exploiting strong winds channeled through the Palghat/Aralvaimozhi gaps.

Answer: (B)

Q15. In the post-independence period, why did the sugar manufacturing industry witness a prominent geographic shift from North India toward Peninsular India?

(A) Peninsular states have higher market demand and lower rail freight charges
(B) Peninsular tropical maritime climate enables higher sucrose content and longer crushing seasons
(C) North Indian sugarcane varieties are more susceptible to winter pest infestations
(D) Labor costs in peninsular sugarcane belts are significantly lower

Sol. Q15. Peninsular India's maritime tropical climate provides a longer crushing season and prevents cane desiccation, resulting in markedly higher sucrose recovery rates.

Answer: (B)